OSITO DRONE STEM ACADEMY

SERIES 09: CODE-BY-NUMBER GRIDS & FLIGHT-PATH MAZES - AGES 7-10 (GRADES 2-4)

THIS STEM COLORING BOOK BELONGS TO

YOUNG PILOT'S NAME / NOMBRE DEL PILOTO

MY DRONE'S CALLSIGN / NOMBRE DE MI DRON

CO-PILOT / GROWN-UP CO-PILOT NAME

Title: Vol. 406: Zara's Looping Command Sequence Math at Backyard Treehouse Lookout

Featured STEM Pilot: Zara (Young Sky Explorer in Lavender Flight Hijab)

Flight Environment: Backyard Treehouse Lookout - Target Age: Ages 7-10 (Grades 2-4)

Core STEM & Math Focus: Repeat 4 Times: Fly Forward and Turn 90 Degrees

Includes: 8 Single-Sided Coloring Plates, 4 Vector STEM Activities, Flight Log & Certificate!

Explore Youth Drone STEM Camps, 3D Printing & Flight Simulators at ositodrone.com

Published by Osito Drone LLC - All Rights Reserved

OFFICIAL YOUTH DRONE SAFETY RULES & FAA TRUST NOTICE

SAFETY RULES FOR RECREATIONAL DRONE PILOTS

1

1. ALWAYS FLY WITH A GROWN-UP

Make sure a parent, teacher, or coach supervises every flight.

2

2. FLY IN OPEN, SAFE SPACES

Stay clear of people, pets, roads, trees, and active airport airspace.

3

3. KEEP DIRECT LINE-OF-SIGHT

Always keep your quadcopter where you can see it with your own eyes.

4

4. USE DUCTED PROPELLER GUARDS

Inspect all 4 ducted prop guards and battery latch before takeoff.

5

5. COUNT & CHECK PRE-FLIGHT MATH

4 propellers + 1 charged battery + 1 adult co-pilot = Safe Takeoff!

6

6. TAKE THE FREE FAA TRUST TEST

Learn official recreational sky rules at faa.gov/uas/trust with family.

OSITO DRONE LLC – PUBLISHING & LEGAL ATTRIBUTION

Osito Drone STEM Academy Curriculum Series - KDP Paperback Edition

Copyright (C) 2026 Osito Drone LLC. All rights reserved.

Published by Osito Drone LLC

Printed in the United States of America - Amazon KDP Print-on-Demand

Standard Edition: 8.5 x 11.0 in, 55# Uncoated White Interior Paper

Publisher Contact: legal@ositodrone.com - Website: ositodrone.com

All characters, curriculum activities, and airframe schematics are property of Osito Drone LLC.

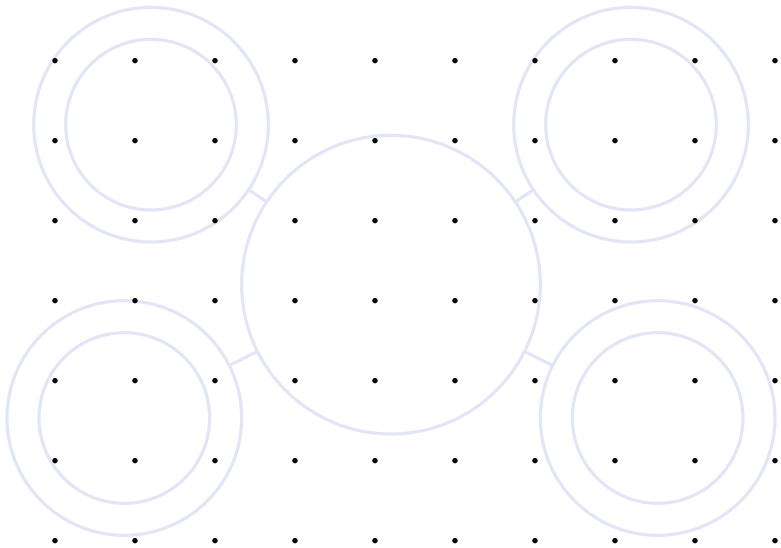
Designed to comply with FAA TRUST youth aviation educational benchmarks.

No part of this publication may be reproduced without prior written permission of Osito Drone LLC,

except for non-commercial educational classroom instruction by licensed educators.

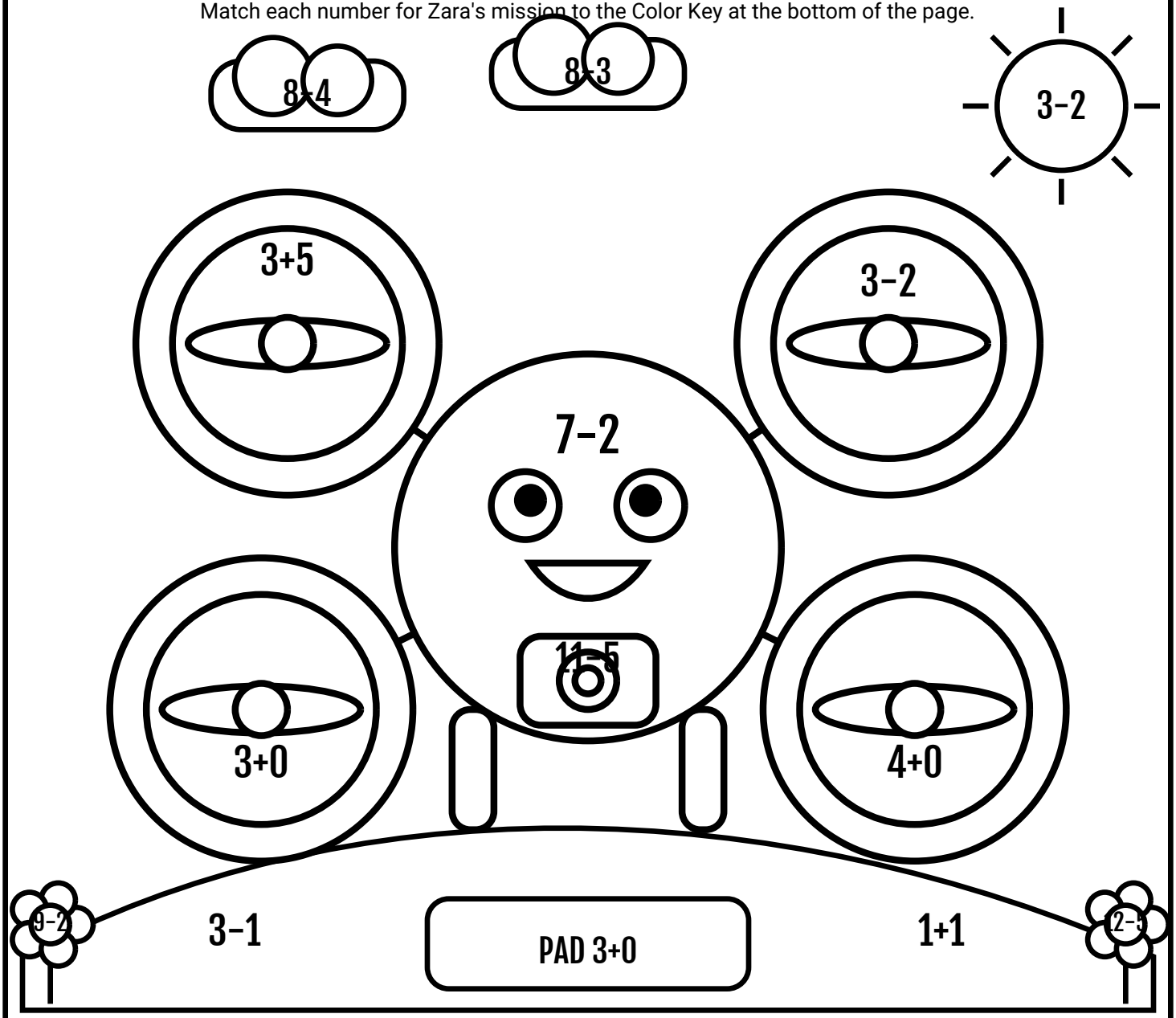


ZARA & PET DRONE – LOOPING COMMAND SEQUENCE MATH



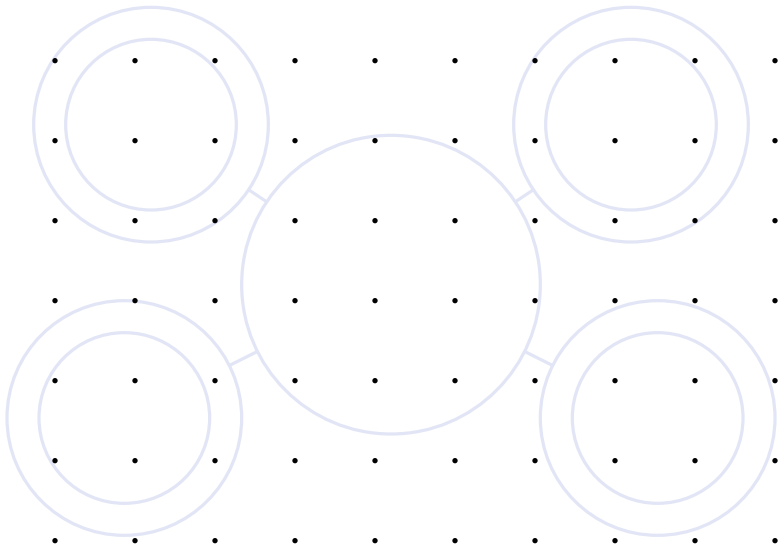
COLOR BY NUMBER: ZARA'S LOOPING COMMAND SEQUENCE MATH

Match each number for Zara's mission to the Color Key at the bottom of the page.



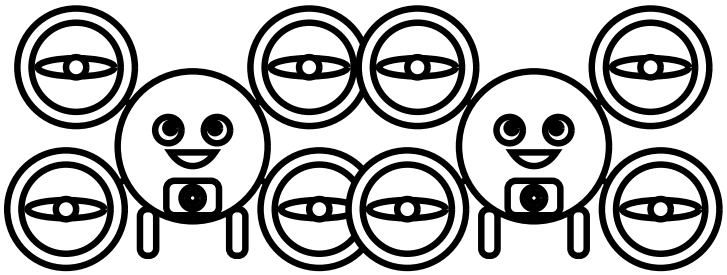
NUMBER-BY-NUMBER COLOR KEY

1 = RED	2 = BLUE	3 = YELLOW	4 = GREEN
5 = ORANGE	6 = PURPLE	7 = TEAL	8 = BROWN



DRONE MATH: ZARA'S LOOPING COMMAND SEQUENCE MATH

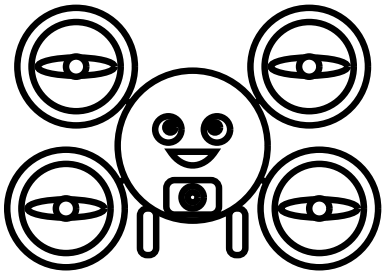
Every quadcopter has 4 ducted propellers! Solve Zara's flight problems and trace the numbers.



MISSION #1: MULTIPLY x 4

2 Drones x 4 Rotors

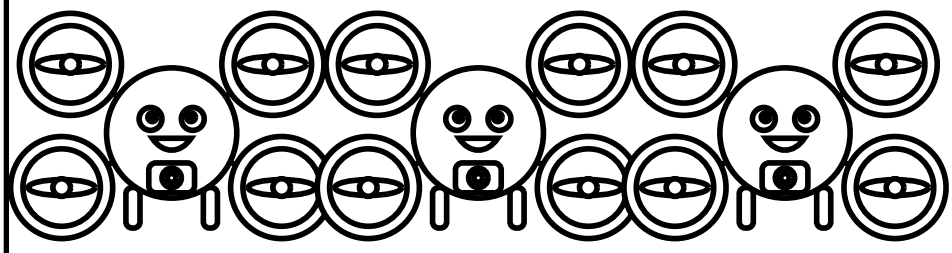
Color the drones & trace numbers!



MISSION #2: BATTERY MATH

15m - 2m Flown

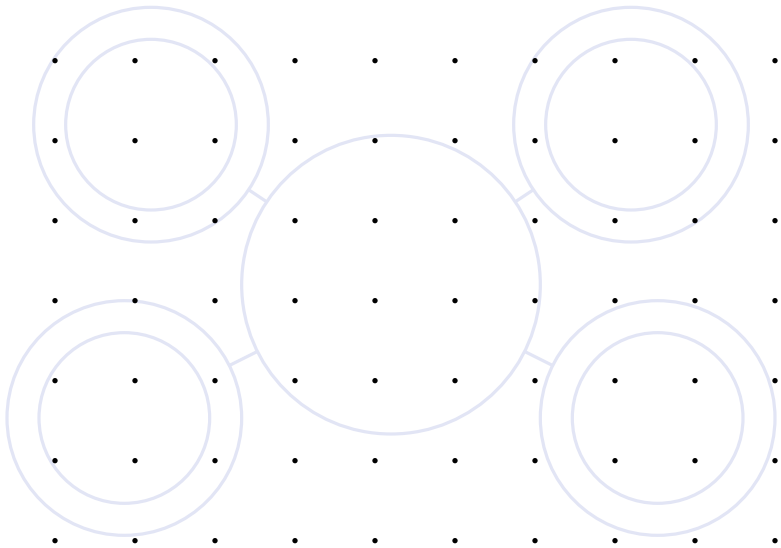
Color the drones & trace numbers!

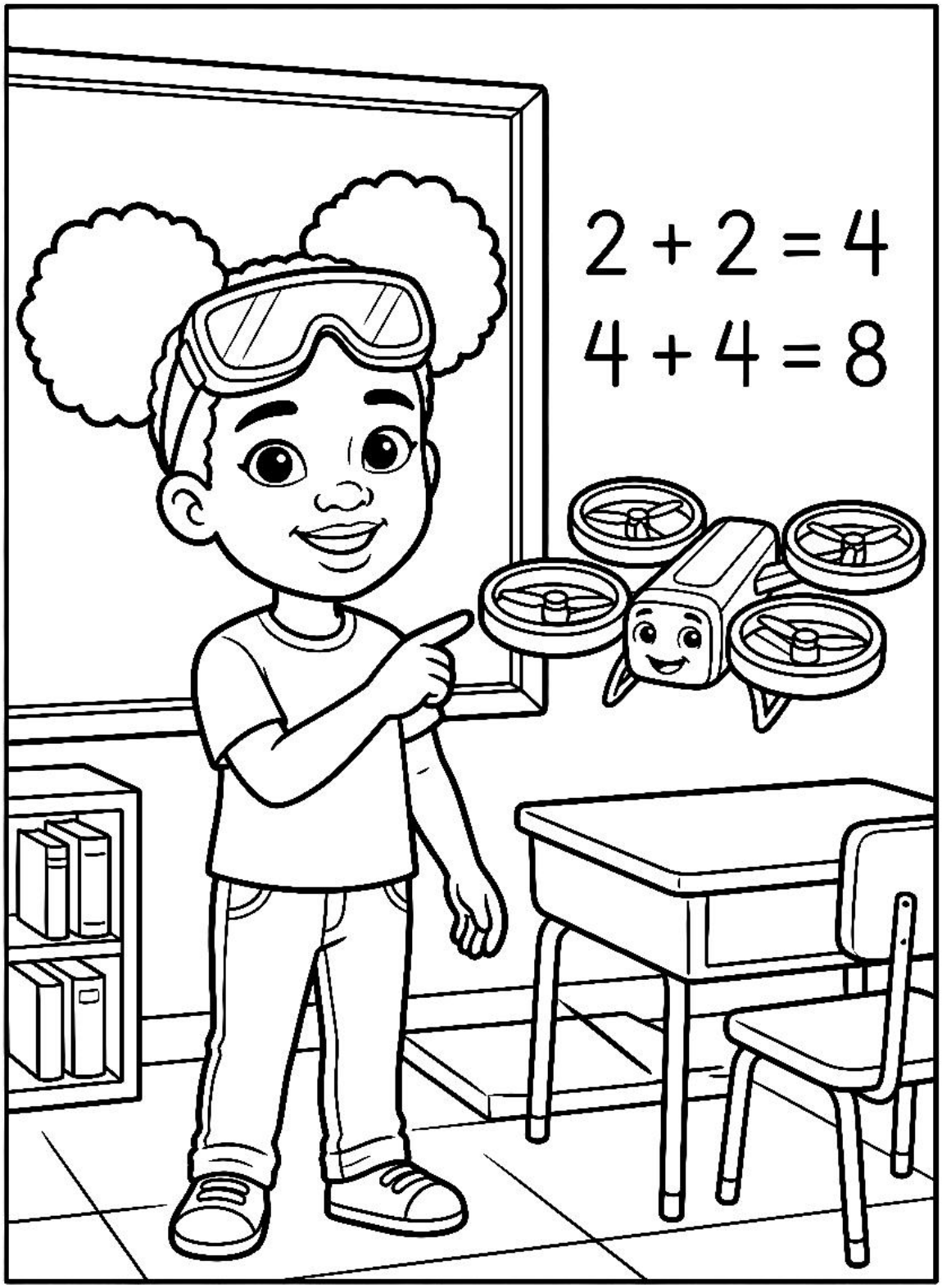


MISSION #3: MULTIPLY x 4

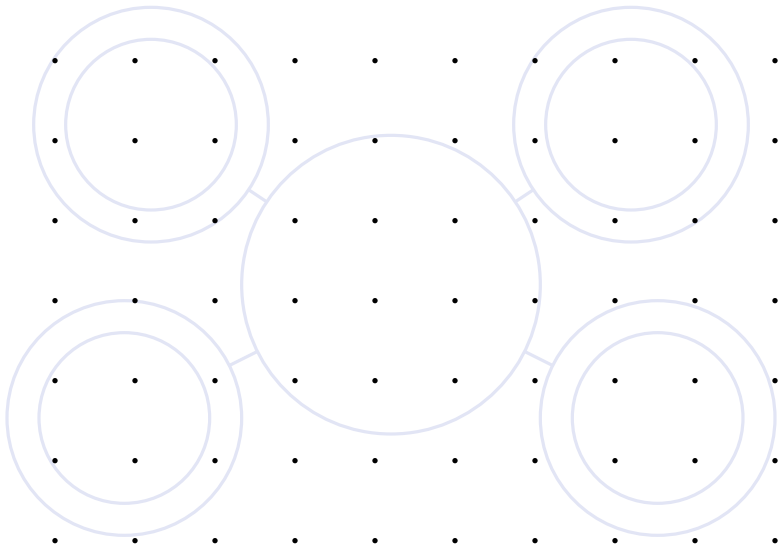
3 Drones x 4 Rotors

Color the drones & trace numbers!



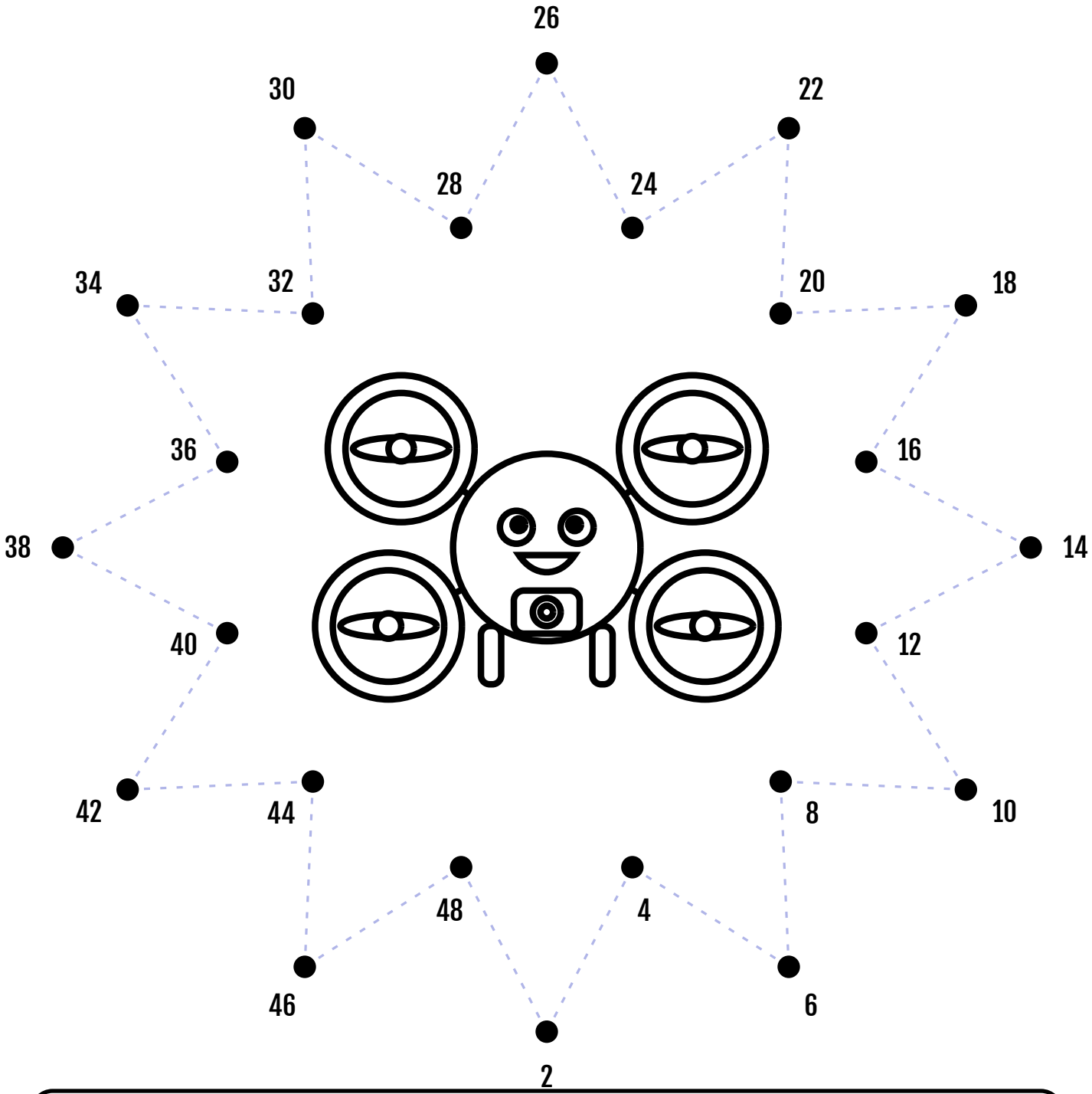


COLOR BY NUMBER & COUNT ROTORS WITH ZARA!

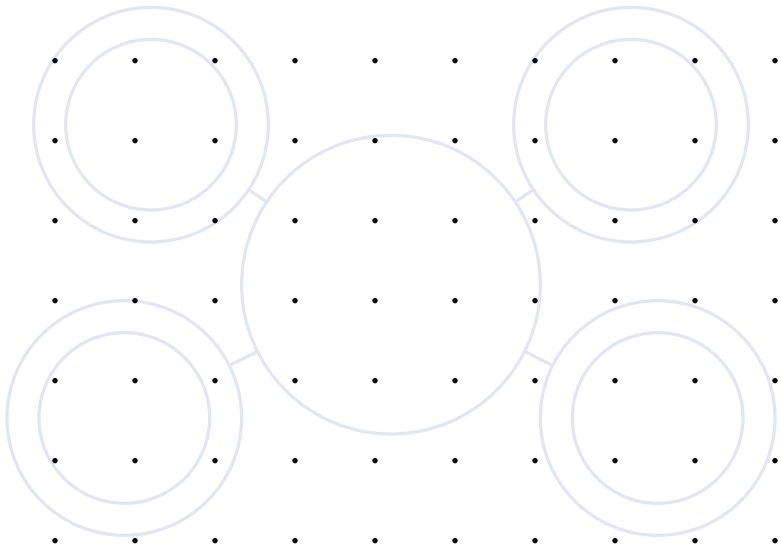


CONNECT THE DOTS: ZARA'S LOOPING COMMAND SEQUENCE MATH

Connect the numbered dots in order counting by 2s to complete the aerospace blueprint!



BONUS FOR ZARA: You connected 24 dots counting by 2s! Final number = 48!



CODE-BY-NUMBER: SOLVE & COLOR PIXEL DRONE

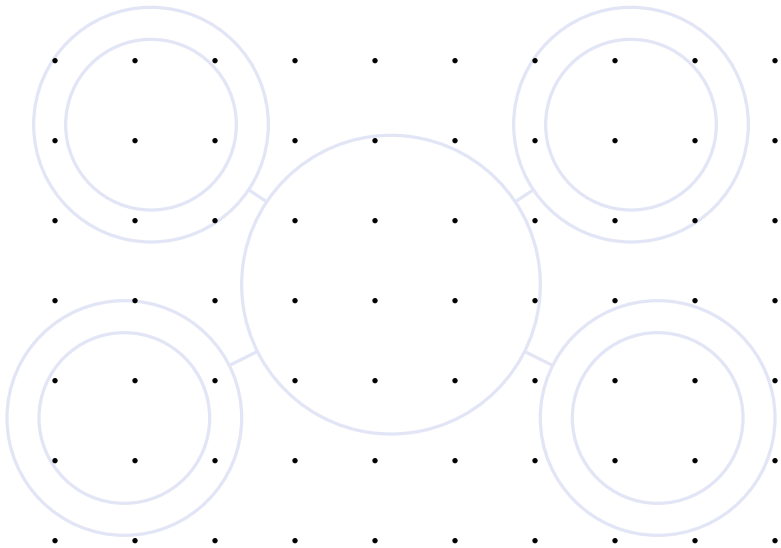
Solve or match the number in each grid square (A1-G7) using the Color Code Key below to reveal pixel art!

	1	2	3	4	5	6	7
A	6-2	1+1	2+0	1+1	2+0	2+0	2+2
B	2+0	2+3	1+0	6-1	1+0	5+0	2+0
C	5-3	2+3	1+0	6-1	4-3	2+3	7-5
D	6-4	10-5	2-1	6-5	3-2	7-2	6-4
E	1+1	10-5	1+0	5+0	1+0	8-3	4-2
F	3-1	1+4	1+0	7-2	4-3	10-5	4-2
G	1+3	3-1	5-3	1+1	1+1	1+1	8-4

PIXEL CODE COLOR KEY

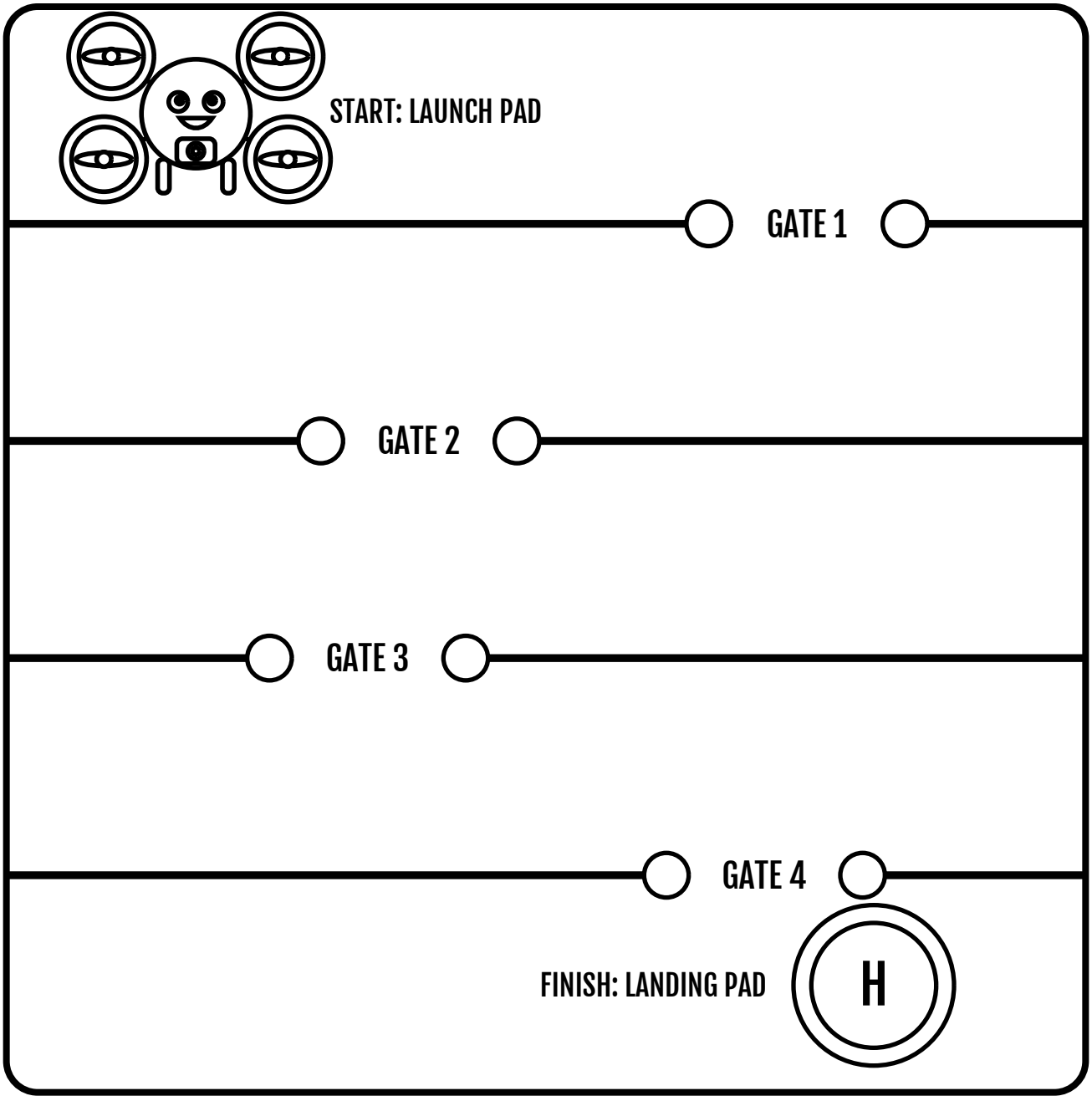
1 = RED 2 = BLUE / INDIGO 3 = GOLD / YELLOW 4 = GREEN 5 = LIGHT SKY BLUE

STEM Tip: Computers and drone cameras store pictures as grids of numbered pixels just like this!

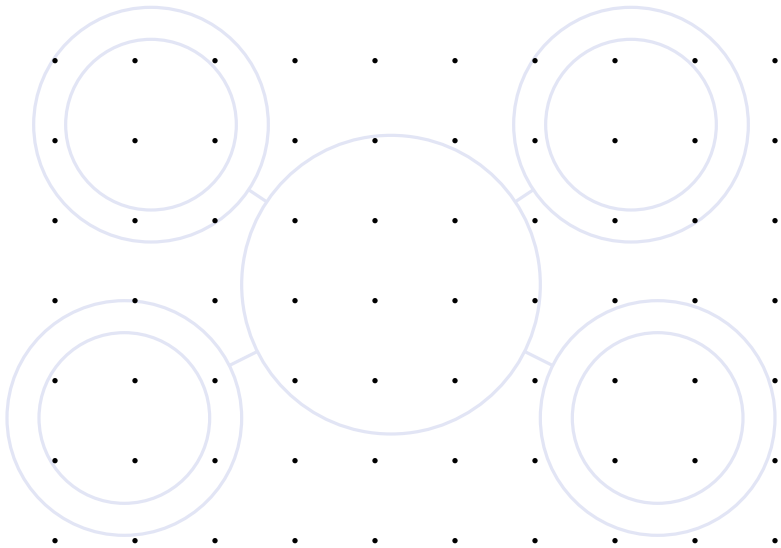


SAFE FLIGHT-PATH MAZE: GUIDE ZARA'S DRONE!

Draw a line from LAUNCH at top-left through open gates in order (1 -- 2 -- 3 -- 4) to LANDING PAD 'H'!

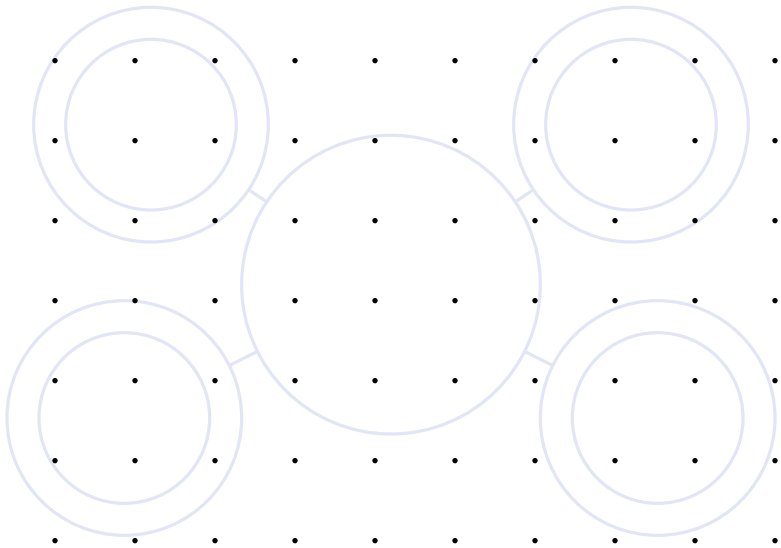


COLORING CHALLENGE: Color all 4 Gate Hoops different colors and draw arrows along your safe flight path!



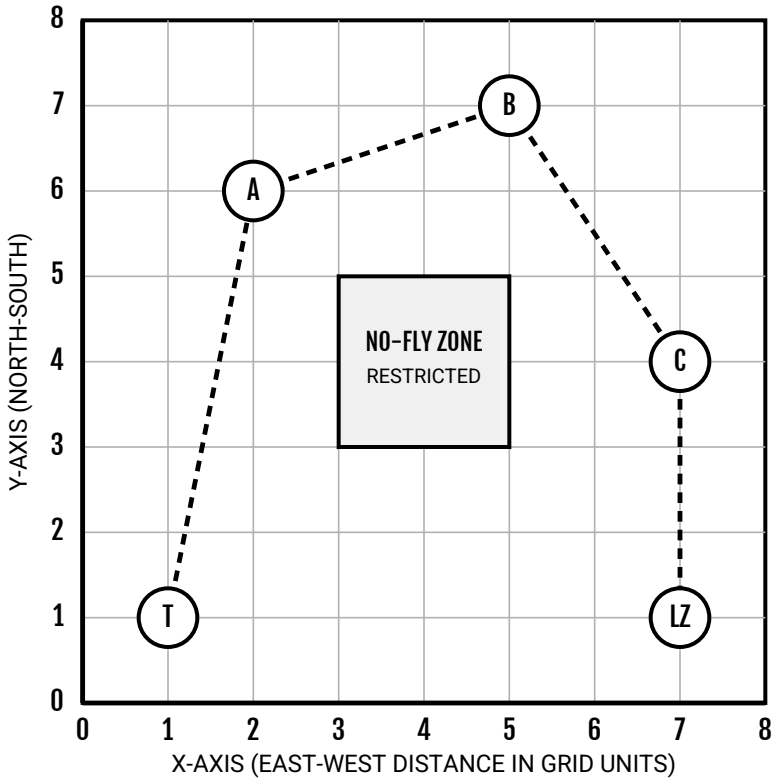


SAFE DRONE FLIGHT AT BACKYARD TREEHOUSE LOOKOUT



COORDINATE FLIGHT PLAN: ZARA'S WAYPOINT NAVIGATION

Plot waypoints on the (X, Y) grid for Zara's Looping Command Sequence Math!

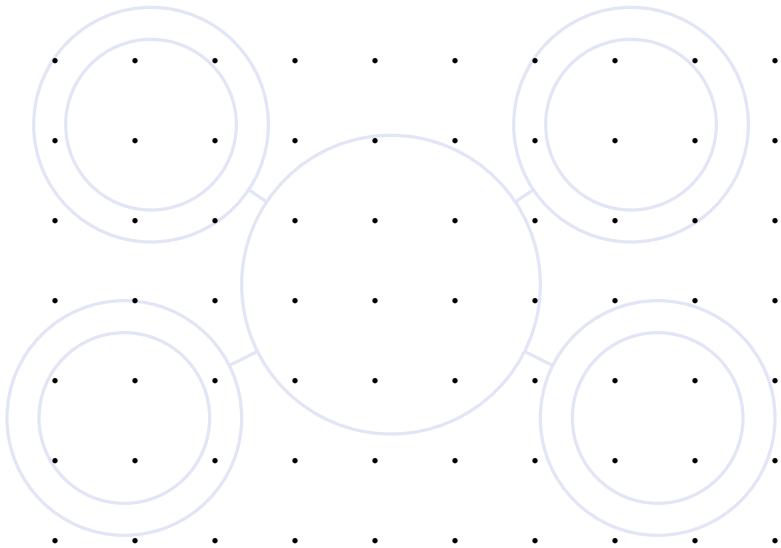


FLIGHT PLAN WAYPOINTS

WP T:	Takeoff Pad	(1 , 1)
WP A:	Zara Survey	(2 , 6)
WP B:	Waypoint Looping Comm	(5 , 7)
WP C:	Bridge Check	(7 , 4)
WP LZ:	Landing Zone	(7 , 1)

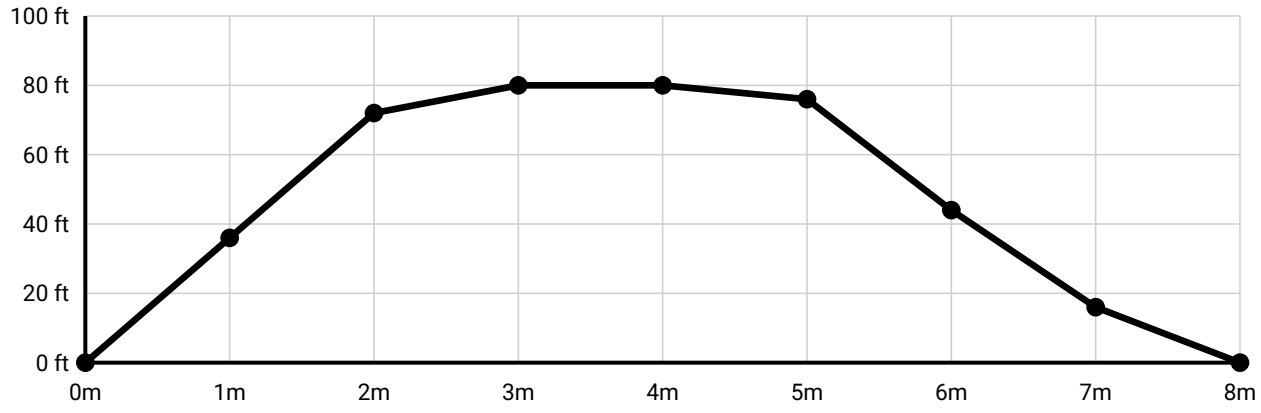
COORDINATE CHALLENGE:

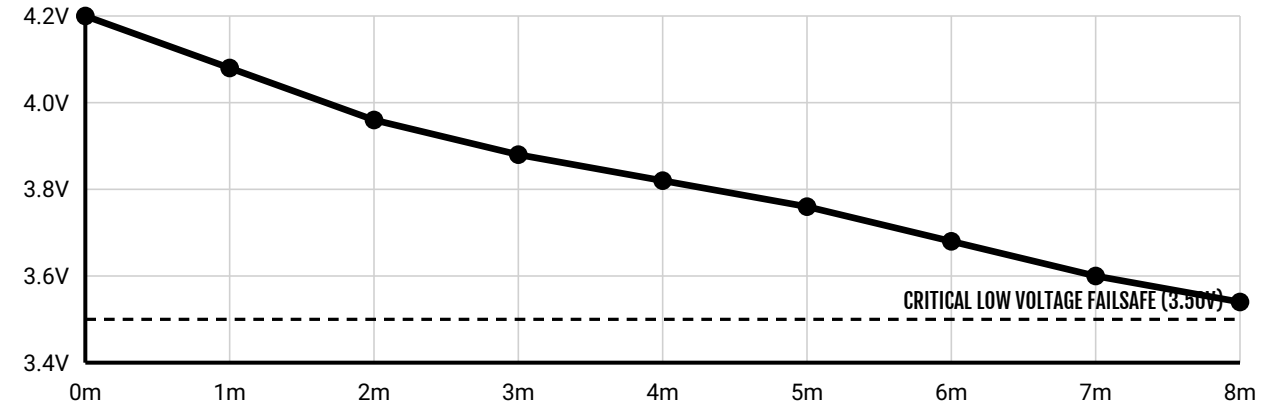
1. Waypoint B coordinates: (5 , 7)
2. Did flight avoid No-Fly Zone? [X] YES
3. Total flight legs flown: 4 legs

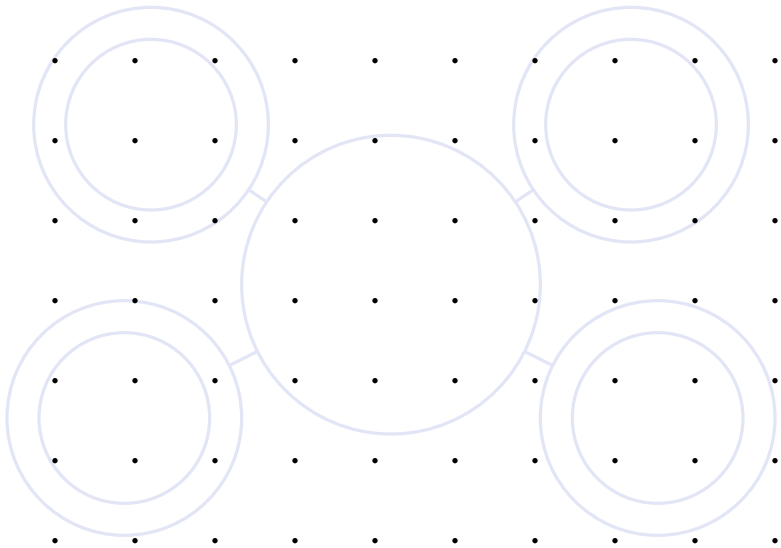


FLIGHT TELEMETRY: ALTITUDE & BATTERY DISCHARGE ENVELOPE

Analyze real flight telemetry data downloaded from the onboard BlackBox flight recorder.

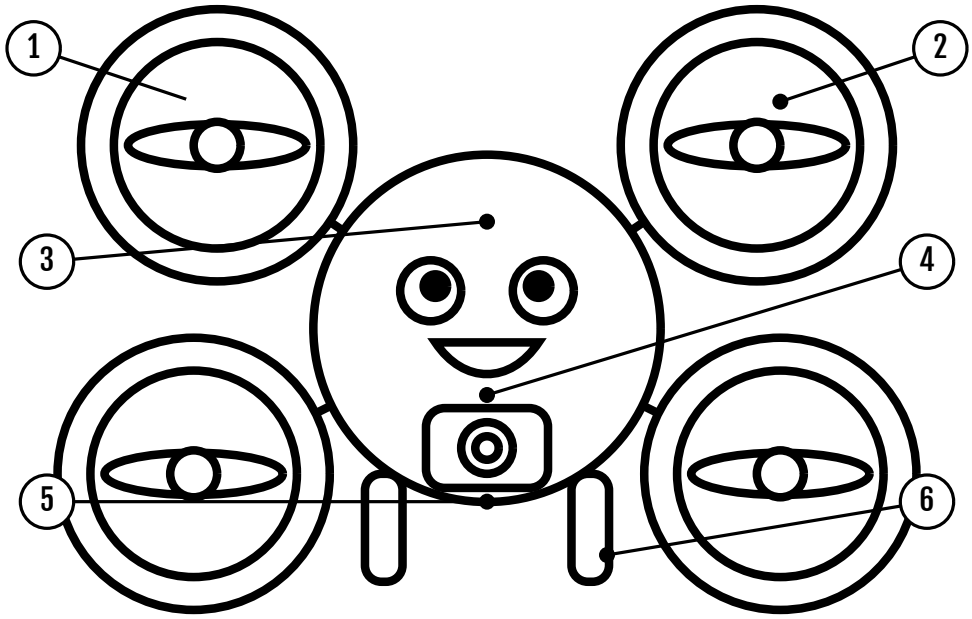






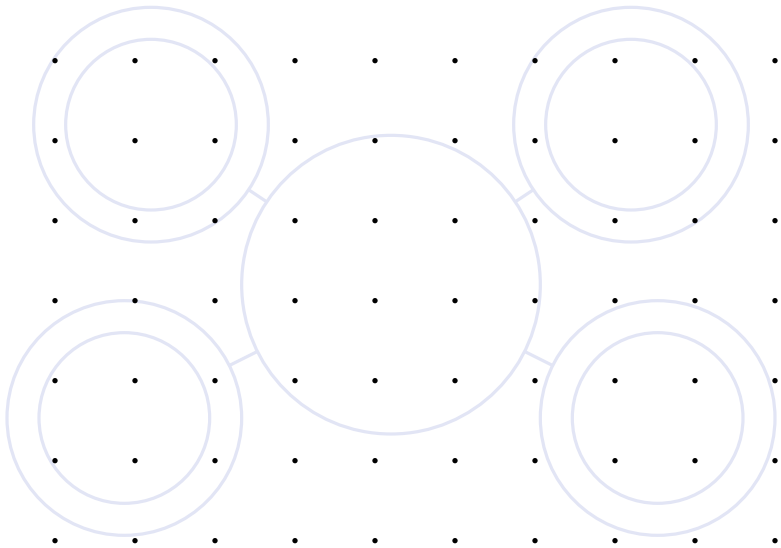
AEROSPACE BLUEPRINT: CUB-75 AVIONICS & AIRFRAME ANATOMY

Study the mechanical airframe and avionics architecture of the Osito CUB-75 STEM training drone.



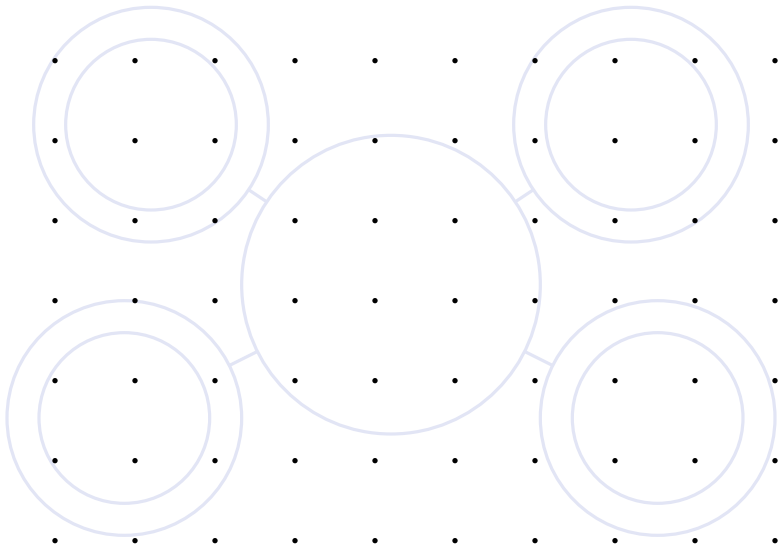
1	Ducted Prop Guards	Duct rings that shield propellers and provide 360-degree impact safety.
2	Brushless DC Motors	0802 19000KV motors that spin props at 30,000 RPM to produce vertical lift.
3	Flight Controller (FC)	Microprocessor & gyroscope that balances the drone 1,000 times every second.
4	4-in-1 ESC Board	Electronic Speed Controller regulating power sent from battery to each motor.
5	Micro FPV Camera	Ultra-wide lens sensor broadcasting real-time video to pilot FPV monitors.
6	1S LiPo Battery	3.8V High-Voltage Lithium Polymer rechargeable cell providing flight energy.

CUB-75 SPECS: Wheelbase: 75mm - AUW Weight: 39.5g - Motors: 0802 19000KV - Flight Time: 8.0 min



FLIGHT PHYSICS: THRUST-TO-WEIGHT & BATTERY ENDURANCE MATH

Apply real aeronautical engineering equations to calculate multirotor performance and endurance.



OFFICIAL OSITO JUNIOR PILOT FLIGHT MISSION LOG

RECORD YOUR ACTUAL FLIGHT MISSIONS AND COMPLETED STEM CHALLENGES

FLIGHT #	DATE	FLIGHT DURATION	MISSION OBJECTIVE	COACH SIGNATURE
Mission 1				
Mission 2				
Mission 3				

OSITO STEM PILOT CERTIFICATE OF ACHIEVEMENT

THIS CERTIFIES THAT JUNIOR PILOT

PILOT NAME HERE

HAS COMPLETED ALL STEM ACTIVITIES AND SAFETY PROTOCOLS IN

BOOK #406: LOOPING COMMAND SEQUENCE MATH

Core Aeronautical Skill: Repeat 4 Times: Fly Forward and Turn 90 Degrees

AND IS OFFICIALLY RECOGNIZED AS A CERTIFIED JUNIOR OSITO DRONE PILOT!

JUNIOR PILOT SIGNATURE

COACH / PARENT SIGNATURE & DATE

